



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

Department of Information Technology (Bsc.IT Semester III)
SCALA FOR D.S PRACTICAL 28

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Class : SYIT	Batch : 1
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PRACTICAL 28

Question:

Compute the Euclidean distance between two Breeze vectors. Use it for nearest neighbor classification.

CODE:

```
build.sbt  Scala.scala ×
1  import breeze.linalg._
2
3  ▶ object Scala extends App
4  {
5      val A = DenseVector(2.0, 3.0)
6      val B = DenseVector(5.0, 7.0)
7
8      val distance = norm(A - B)
9
10     println("Euclidean Distance: " + distance)
11
12     val C = DenseVector(2.5, 3.5)
13
14     val d1 = norm(C - A)
15     val d2 = norm(C - B)
16
17     if (d1 < d2)
18     |     println("Nearest Neighbor: A")
19     else
20     |     println("Nearest Neighbor: B")
21 }
```

OUTPUT:

```
Euclidean Distance: 5.0
Nearest Neighbor: A

Process finished with exit code 0
```